

In [1]:

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
```

In [2]:

```
dataset=pd.read_csv("Social_Network_Ads.csv")
```

In [3]:

```
dataset
```

Out[3]:

	User ID	Gender	Age	EstimatedSalary	Purchased
0	15624510	Male	19	19000	0
1	15810944	Male	35	20000	0
2	15668575	Female	26	43000	0
3	15603246	Female	27	57000	0
4	15804002	Male	19	76000	0
...	...	...	...	...	...
395	15691863	Female	46	41000	1
396	15706071	Male	51	23000	1
397	15654296	Female	50	20000	1
398	15755018	Male	36	33000	0
399	15594041	Female	49	36000	1

400 rows × 5 columns

In [4]:

```
dataset = dataset.drop(columns=['User ID'])
```

In [5]:

```
dataset=pd.get_dummies(dataset,drop_first=True)
```

In [6]:

```
dataset
```

Out[6]:

	Age	EstimatedSalary	Purchased	Gender_Male
0	19	19000	0	True
1	35	20000	0	True
2	26	43000	0	False
3	27	57000	0	False
4	19	76000	0	True
...	...	...	...	...

	Age	EstimatedSalary	Purchased	Gender_Male
395	46	41000	1	False
396	51	23000	1	True
397	50	20000	1	False
398	36	33000	0	True
399	49	36000	1	False

400 rows × 4 columns

In [7]:

```
dataset.columns
```

Out[7]:

```
Index(['Age', 'EstimatedSalary', 'Purchased', 'Gender_Male'], dtype='object')
```

In [8]:

```
x=dataset[['Age', 'EstimatedSalary', 'Gender_Male']]
y=dataset[['Purchased']]
```

In [9]:

```
from sklearn.model_selection import train_test_split
x_train,x_test,y_train,y_test=train_test_split(x,y,test_size=0.2,random_state=11)
```

In [10]:

```
from sklearn.ensemble import RandomForestClassifier
classifier = RandomForestClassifier(random_state=0, n_estimators=100,criterion='entropy')
classifier.fit(x, y)
```

C:\Users\Gayathri R\AppData\Local\Programs\Python\Python312\Lib\site-packages\sklearn\base.py:1474: DataConversionWarning: A column-vector y was passed when a 1d array was expected. Please change the shape of y to (n_samples,), for example using ravel().
return fit_method(estimator, *args, **kwargs)

Out[10]:

```
RandomForestClassifier
RandomForestClassifier(criterion='entropy', random_state=0)
```

In [11]:

```
y_pred=classifier.predict(x_test)
```

In [12]:

```
from sklearn.metrics import confusion_matrix
cm=confusion_matrix(y_test,y_pred)
```

In [13]:

```
print(cm)
```

```
[[53  0]
 [ 0 27]]
```

In [16]:

```
from sklearn.metrics import classification_report
clf_report=classification_report(y_test,y_pred)
```

In [17]:

```
print(clf_report)
```

	precision	recall	f1-score	support
0	1.00	1.00	1.00	53
1	1.00	1.00	1.00	27
accuracy			1.00	80
macro avg	1.00	1.00	1.00	80
weighted avg	1.00	1.00	1.00	80

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```

In [14]:

```
from sklearn.tree import DecisionTreeClassifier
clf = DecisionTreeClassifier(random_state=100)
clf.fit(x, y)
```

Out[14]:

```
DecisionTreeClassifier
DecisionTreeClassifier(random_state=0)
```

In [16]:

```
y_pred=clf.predict(x_test)
```

In [17]:

```
from sklearn.metrics import confusion_matrix
cm=confusion_matrix(y_test,y_pred)
```

In [18]:

```
print(cm)
```

```
[[53  0]
 [ 1 26]]
```

In [19]:

```
from sklearn.metrics import classification_report
clf_report=classification_report(y_test,y_pred)
```

In [20]:

```
print(clf_report)
```

	precision	recall	f1-score	support
0	0.98	1.00	0.99	53
1	1.00	0.96	0.98	27
accuracy			0.99	80
macro avg	0.99	0.98	0.99	80
weighted avg	0.99	0.99	0.99	80

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2	26	43000	0	False
3	27	57000	0	False
4	19	76000	0	True
...	...	...	...	...

	Age	EstimatedSalary	Purchased	Gender_Male
395	46	41000	1	False
396	51	23000	1	True
397	50	20000	1	False
398	36	33000	0	True
399	49	36000	1	False

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In [7]:

```
x=dataset[['Age', 'EstimatedSalary', 'Gender_Male']]
y=dataset[['Purchased']]
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In [8]:

```
from sklearn.model_selection import train_test_split
x_train,x_test,y_train,y_test=train_test_split(x,y,test_size=0.2,random_state=11)
```

In [9]:

```
from sklearn.preprocessing import StandardScaler
sc=StandardScaler()
x_train=sc.fit_transform(x_train)
x_test=sc.transform(x_test)
```

In [10]:

```
from sklearn.svm import SVC
clf=SVC(kernel="poly",C=100)
cl=clf.fit(x_train,y_train)
```

C:\Users\Gayathri R\AppData\Local\Programs\Python\Python312\Lib\site-packages\sklearn\utils\validation.py:1300: DataConversionWarning: A column-vector y was passed when a 1d array was expected. Please change the shape of y to (n_samples,), for example using ravel().

```
y = column_or_1d(y, warn=True)
```

In [12]:

```
y_pred=clf.predict(x_test)
y_pred
```

Out[12]:

```
array([0, 1, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
        0, 0, 0, 0, 0, 1, 0, 0, 1, 0, 0, 0, 0, 1, 0, 0, 0, 1, 0, 0, 0, 0,
        1, 1, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 1, 0, 1, 0, 0, 1, 0, 1, 0, 1, 0,
        1, 1, 0, 0, 1, 1, 0, 0, 0, 0, 1, 1, 0, 0], dtype=int64)
```

In [13]:

```
from sklearn.metrics import confusion_matrix
cm=confusion_matrix(y_test,y_pred)
```

In [14]:

```
print(cm)
```

```
[[49  4]
 [11 16]]
```

In [15]:


```
from sklearn.metrics import classification_report
clf_report=classification_report(y_test,y_pred)
```

In [16]:

```
print(clf_report)
```

	precision	recall	f1-score	support
0	0.82	0.92	0.87	53
1	0.80	0.59	0.68	27
accuracy			0.81	80
macro avg	0.81	0.76	0.77	80
weighted avg	0.81	0.81	0.80	80

In []: